# Field Check Guide: Potato After the Harvest

## আলু রাখবেন না বেচবেন: কৃষক আর হিমাগারের কাছে প্রশ্নমালা, বগুড়া

**Why we are doing this.** KrishiVault tells a potato farmer whether to sell now, store until a month, or sell part and store the rest. The sums use the cold-storage rent, the sack size, weight loss, carrying and selling costs, and loan interest. Some of these come from our field notes; **many are still our guesses**. This check replaces the guesses with real Bogura numbers, and tells us when and why farmers sell.

> **The golden rule.** Only real answers from real people go into the engine or onto a slide. Never invent a number or a quote. If a judge asks "how many farmers and cold storages did you talk to, and where?", we answer with what we really did.

## Who to talk to

- **At least 5 potato farmers** (10 is better) in **Shibganj and Sariakandi**, the pilot upazilas.
- Mix: farmers who stored in a cold storage last season and farmers who sold at the harvest; small and larger; some women farmers.
- **2 cold storages**: the manager or owner (page 5).
- About 25 minutes a farmer, 20 a cold storage. Two of us: one asks, one writes.

**Good timing:** in October the potatoes stored in March are still in the cold storages, so farmers remember the rent, the advance and the weight they put in.

## How to ask

- **Ask an open question first**, then ask for the number.
- Don't suggest answers. Ask "why did you sell then?", not "did you sell because of dadon?"
- Use their units: মণ, বিঘা, বস্তা, কেজি. Write the unit with every number.
- **Write numbers, not impressions.** "About ৳250 a sack" beats "cheap".
- Write exact words for anything that could be a quote.
- Ask before every photo or recording; mark it on the form.

## Before you start: read this aloud

"আমরা ছাত্র, একটা কৃষি প্রকল্পের জন্য আপনার অভিজ্ঞতা জানতে চাই। আপনার নাম কোথাও প্রকাশ হবে না। ছবি বা রেকর্ড শুধু আপনার অনুমতি নিয়ে। আমরা কোনো টাকা, ঋণ বা সাহায্যের প্রতিশ্রুতি দিচ্ছি না। কথা বলবেন?"

We are students and want to learn from your experience for a farming project. Your name will not be published. Photos or recordings only with your permission. We are not promising money, loans or help. Will you talk to us?

**To cold storages:** say the same, and say clearly that KrishiVault has **no partners yet**; the cold storages, banks and buyers in our prototype are made up (demo). We are collecting information, not signing anyone.

## How this guide is laid out

- **Pages 2 to 4:** farmer questions in seven groups, and how to play the call to a farmer. ★ marks a must-ask question. "Checks" names the number in KrishiVault's engine that the answer replaces.
- **Page 5:** cold-storage questions.
- **Pages 6 and 7:** answer forms; print one per farmer and one per cold storage.
- **Page 8:** the tally sheet, and the table that updates the engine.
- **Page 9:** turning answers into slides, and what we expect to hear.

# Farmer questions (1 of 3)

★ must ask · Ask in Bangla, in the farmer's own words · "Checks" = the engine number or plan claim the answer tests (page 8 lists them all)

## A. The farm and the phone · the quantity, and why we use calls

| # | ASK IN BANGLA | MEANING | CHECKS |
|---|---|---|---|
| 1 | গত মৌসুমে কত বিঘা জমিতে আলু করেছিলেন? মোট কত মণ উঠেছিল? | How many bigha of potato last season? How many maunds harvested? | The quantity a farmer gives in the call; maunds per bigha |
| 2 ★ | আপনার ফোনটা কেমন, বাটনওয়ালা না টাচ? মেসেজ (SMS) নিজে পড়েন, নাকি কেউ পড়ে দেয়? | Button or touch phone? Do you read SMS yourself, or does someone read it? | Voice call first, SMS as a receipt |

## B. Selling last season · when, at what price, and why

| # | ASK IN BANGLA | MEANING | CHECKS |
|---|---|---|---|
| 3 ★ | গত মৌসুমের আলু কবে কবে, কত মণ করে বেচলেন? কেজি বা মণ কত টাকায়? | When did you sell, how many maunds each time, at what price per kg or maund? | Farm-gate prices by month: the engine now assumes they move with town retail prices |
| 4 ★ | তখনই বেচলেন কেন? টাকার দরকার, দাদন, রাখার জায়গা নেই, দাম পড়ার ভয়, নাকি পচার ভয়? | Why sell then? Cash need, dadon, no storage, fear of a price fall, or rot? | The trap: "farmers sell within 1 to 3 months, afraid of a fall" |
| 5 | কার কাছে বেচলেন: ফড়িয়া, আড়তদার, নাকি নিজে হাটে? দামটা কে ঠিক করে? | Who bought: a faria, an aratdar, or you at the haat? Who set the price? | Market access: buyers ranked by money in hand |
| 6 | মাঠ থেকে বেচলে বস্তায় কত খরচ পড়ে: বস্তা, শ্রমিক, ভ্যান? | Selling at the field: cost per sack (sack, labour, van)? | Selling at the field: ৳10 a sack (guess) |
| 7 | দাদন বা আগাম টাকা নিয়েছিলেন? কার কাছ থেকে, কত? তার কাছেই বেচতে হয়েছে? | Did you take dadon? From whom, how much? Did you have to sell to him? | Forced sales; the dadon question in the call |
| 8 | বেচার পরে বাজারে আলুর দাম কত উঠেছিল, কোন মাসে? | How high did the market price go later, and in which month? | The price rise farmers miss |

## C. Cold storage · ask farmers who stored; others go to question 16

| # | ASK IN BANGLA | MEANING | CHECKS |
|---|---|---|---|
| 9 ★ | গত মৌসুমে হিমাগারে কত বস্তা রেখেছিলেন? খাওয়ার আলু, না বীজ? | How many sacks did you store? Potatoes to eat (sell), or seed? | "Farmers mostly store only seed" |
| 10 ★ | এক বস্তায় কত কেজি? ভাড়া বস্তায় কত, নাকি কেজিতে? ১ মাস রাখলেও কি একই ভাড়া? কবে দিতে হয়? | Kg in a sack? Rent per sack, or per kg? The same for 1 month as for 9? When paid? | 84 kg sack, ৳250 rent (field notes); the flat rent |

# Farmer questions (2 of 3)

## C. Cold storage, continued

| # | ASK IN BANGLA | MEANING | CHECKS |
|---|---|---|---|
| 11 | খালি বস্তা কত টাকা? জমি থেকে হিমাগারে নিতে বস্তায় কত? ওঠানো-নামানোর মজুরি কত? | An empty sack? Carrying to the cold storage, per sack? Loading and unloading labour? | Sack ৳40, carrying ৳25, loading ৳15 (guesses) |
| 12 ★ | হিমাগার থেকে বস্তা জামানত রেখে টাকা (অগ্রিম) নিয়েছিলেন? বস্তায় কত? সুদ বা খরচ কত? কবে শোধ করতে হয়? না দিলে কী হয়? | Did you take an advance against the sacks? How much per sack? Interest or charge? When repaid? What if you can't? | Advance ৳250 a sack (notes), 14% a year (guess) |
| 13 ★ | কবে বের করলেন? রাখার সময় বস্তার ওজন কত ছিল, বের করার সময় কত? কত কেজি পচা বাছাই করে ফেলতে হলো? | When did you take them out? Sack weight going in, and coming out? How many kg sorted out as rotten? | Loss 0.5% a month, 2% sorted out (guesses) |
| 14 | হিমাগার থেকে বেচতে বস্তায় কত খরচ: গাড়ি, কমিশন, শ্রমিক? | Selling from the cold storage: cost per sack (transport, commission, labour)? | Selling from store: ৳30 a sack (guess) |
| 15 | ওই মাসে বের করলেন কেন? | Why did you take them out in that month? | Why stored potatoes are sold early |
| 16 | (যাঁরা রাখেননি) হিমাগারে রাখেননি কেন: ভাড়া, জায়গা নেই, অনেক দূর, নাকি তখনই টাকা দরকার? | (Didn't store) Why not: the rent, no space, too far, or cash needed at once? | What stops farmers from storing |

## D. Home storage

| # | ASK IN BANGLA | MEANING | CHECKS |
|---|---|---|---|
| 17 | বাড়িতে আলু কত দিন রাখা যায়? মাসে মণে কত কেজি পচে বা শুকিয়ে যায়? | How long does potato keep at home? How many kg per maund rot or dry out each month? | Home loss 4% a month (guess), 3 months (notes) |

## E. Money and loans

| # | ASK IN BANGLA | MEANING | CHECKS |
|---|---|---|---|
| 18 | আলু তোলার সময় কিসের জন্য টাকার দরকার পড়ে? কত টাকা, কত দিনের মধ্যে? | At the harvest, what do you need cash for? How much, within how many days? | The cash need in the call; "sell part" sizes |
| 19 | ব্যাংক বা এনজিও থেকে কখনো ঋণ নিয়েছেন? সুদ কত ছিল? | Ever borrowed from a bank or NGO? At what interest? | Bank interest 11% a year (guess) |
| 20 ★ | হিমাগারের রসিদ দেখিয়ে ব্যাংক যদি বস্তা জামানতে ঋণ দেয়, নেবেন? কত টাকা দরকার হতো? | If a bank lent against your stored sacks (the cold-storage receipt), would you take it? How much? | Part 4: loans against stored sacks |

## F. Group storage

| # | ASK IN BANGLA | MEANING | CHECKS |
|---|---|---|---|
| 21 | পাশের কয়েকজন মিলে একসাথে হিমাগারে রাখলে বা এক গাড়িতে নিলে রাজি হবেন? দলটা কে চালালে বিশ্বাস করবেন? | Would you store together with neighbours, or share a truck? Whom would you trust to run the group? | Part 6: group storage |

## Farmer questions (3 of 3)

| G. Test the call, and the closing question | | | |
|---|---|---|---|
| # | ASK IN BANGLA | MEANING | CHECKS |
| 22 ★ | (কলের প্রিভিউ চালান, বা উত্তরটা পড়ে শোনান) কী বুঝলেন? এখন কী করবেন? | (Play the call preview, or read the answer aloud.) What did you understand? What would you do now? | Ask them to repeat the advice in their own words |
| 23 | উত্তর দিতে বোতাম চাপা সহজ, নাকি মুখে বলা? এই পরামর্শ বিশ্বাস করবেন? কেন? | Easier to press keys or to speak? Would you trust this advice? Why? | Keypad and voice; trust |
| 24 | কোন খবরটা জানা থাকলে আপনি আলু আরও পরে বেচতেন? | What would you have needed to know to sell later? | What the advice must say |
| 25 ★ | একটাই সমস্যা যদি আমরা সমাধান করতে পারি, কোনটা চাইবেন? | If we could fix just one problem, which one? | The quote for the opening slide |

**Closing:** "আপনার সময়ের জন্য ধন্যবাদ। আপনার কথাগুলো আমাদের সেবা ঠিকভাবে বানাতে সাহায্য করবে।" Check every number on the form before you leave. Confirm what they agreed to (photo, recording, or neither). Give each farmer a code (F01, F02 ...); never put a phone number on a slide.

## How to play the call in the field

**With a laptop** (best): start KrishiVault on it before the visit (`docs/SCREENS.md`), open `/call`, choose the season "Next harvest", press **New call**, then **1**, and type the farmer's own numbers on the keypad, each followed by **#**: maunds, today's price per kg, cash needed (0 if none) and, if it asks, the days; then **1** to confirm. Let the farmer listen, then ask questions 22 and 23. The keypad works without internet; the Bangla voice may need it, so if there is no sound, read the answer from the screen.

**Without a laptop:** read the example answer below aloud, slowly, after saying: "ধরুন একজন কৃষকের ৮০ মণ আলু উঠেছে, ফড়িয়া কেজিতে ১৬ টাকা দিতে চায়, আর ৪ দিনের মধ্যে ২৫ হাজার টাকা দরকার। আমাদের ফোনটা তাকে এটা বলবে:" (Imagine a farmer with 80 maunds, offered ৳16 a kg, who needs ৳25,000 within 4 days. Our call would tell him this:)

EXAMPLE ANSWER, WORD FOR WORD FROM KRISHIVAULT'S ENGINE (A FORECAST FOR THE MARCH 2027 HARVEST, NOT A PROMISE)

"পরামর্শ: এখন ১৯ বস্তা বেচুন, হাতে আসবে প্রায় ২৫,০০০ টাকা। বাকি ১৭ বস্তা হিমাগারে রেখে ডিসেম্বর মাসে বেচুন। সব মিলিয়ে গড়ে প্রায় ৫৬,৫০০ টাকা, এখন সব বেচার চেয়ে প্রায় ৯,১০০ টাকা বেশি; দাম খারাপ হলে প্রায় ৩৮,২০০ টাকা। এতে আপনার ২৫,০০০ টাকার দরকার মিটবে। দাম বাড়বেই এমন কথা নেই, তাই কিছুটা এখনই বেচা নিরাপদ।"

Sell 19 sacks now: about ৳25,000 in hand. Store the other 17 and sell in December. About ৳56,500 in all on average, about ৳9,100 more than selling everything now; about ৳38,200 if prices go badly. That covers your ৳25,000. Prices may not rise, so selling some now is safer.

**What to listen for:** can the farmer say back how many sacks to sell now and until when to store the rest? Did "দাম খারাপ হলে" (if prices go badly) make sense? Write the farmer's exact words.

## Cold-storage questions

The manager or owner of 2 cold storages near Shibganj and Sariakandi. Say that KrishiVault has no partners yet and promises nothing. "Checks" names the engine number the answer replaces.

| # | ASK IN BANGLA | MEANING | CHECKS |
|---|---|---|---|
| C1 | হিমাগারে কত বস্তা ধরে? এ মৌসুমে কত ভরেছে? কবে ভরে, কবে খালি হয়? | How many sacks does it hold? How full this season? When does it fill and empty? | Stores empty by December; 50,000 to 2.5 lakh sacks (notes) |
| C2 ★ | এ মৌসুমে ভাড়া কত: বস্তায় না কেজিতে? এক বস্তায় কত কেজি? ১ মাস আর ৯ মাসে কি একই ভাড়া? কবে দিতে হয়? | This season's rent: per sack or per kg? Kg in a sack? The same for 1 month and 9? When paid? | Rent, sack size, 9-month season |
| C3 | বেশি বস্তা বা কয়েকজন মিলে রাখলে ছাড় দেন? কত বস্তা হলে, কত ছাড়? | A discount for many sacks, or for a group? From how many sacks, how much off? | Group-storage discount tiers (demo figures now) |
| C4 ★ | বস্তা জামানত রেখে কৃষককে অগ্রিম দেন? বস্তায় কত? সুদ বা সার্ভিস চার্জ কত? শোধ না হলে কী করেন? | Do you advance money against the sacks? How much per sack? Interest or service charge? What if it is not repaid? | Advance and its interest |
| C5 ★ | রাখার পর মাসে ওজন কত কমে? বের করার সময় শতকরা কত পচা বা বাদ যায়? | How much weight is lost each month? What share is rotten or sorted out at the exit? | Weight loss and sorting loss |
| C6 | ভাড়া ছাড়া কৃষকের কাছ থেকে আর কী খরচ নেন: ওঠানো-নামানো, বস্তা, সাজানো? | Apart from rent, what else do farmers pay: loading, sacks, stacking? | Loading and sack costs |
| C7 ★ | কৃষককে কী রসিদ দেন? ব্যাংক চাইলে বস্তা আছে কিনা নিশ্চিত করে দেবেন? কোনো ব্যাংক কি আপনার রসিদ দেখে ঋণ দিয়েছে? | What receipt does a farmer get? Would you confirm the sacks to a bank? Has any bank lent against your receipts? | Part 4: the cold storage confirms the sacks, the bank lends |
| C8 | কারা বেশি রাখে, কৃষক না ব্যবসায়ী? খাওয়ার আলু না বীজ? মোটামুটি কত ভাগ? | Who stores more, farmers or traders? Table potato or seed? Roughly what shares? | "Traders store for sale; farmers mostly store seed" |
| C9 | কৃষকেরা সাধারণত কোন মাসে আলু বের করে? ব্যবসায়ীরা কোন মাসে? | In which months do farmers usually take potatoes out? And traders? | "Farmers sell within 1 to 3 months" |
| C10 | মৌসুমের শুরুতে দল বেঁধে বুকিং পেলে আপনার লাভ কী? এমন বুকিংয়ের জন্য কি কোনো ফি দেবেন? | What would early group bookings be worth to you? Would you pay a fee for such a booking? | The business model: a fee per group booking |

**After the visit:** ask for a rate card or a receipt to photograph (with permission): a printed rate beats a remembered one. Note the cold storage's name and upazila on the form, but on slides write only "a cold storage in Shibganj".

ANSWER FORM · PRINT ONE PER FARMER

# Farmer কৃষক F ____

Date

Upazila / village

Interviewers

Consent given
☐ Yes ☐ No

Photo allowed
☐ Yes ☐ No

Recording allowed
☐ Yes ☐ No

## A · FARM AND PHONE

1 · Potato land (bigha)

1 · Harvest (maunds)

2 · Phone · reads SMS
☐ Button ☐ Touch ☐ Self
☐ Someone

## B · SELLING LAST SEASON

3 · Each sale: month · maunds · ৳ per kg or per maund (write the unit)

4 · Why sold then
☐ Cash ☐ Dadon ☐ No storage ☐ Fear of price fall ☐ Rot ☐ Other

5 · Sold to / who set the price

6 · Field sale cost ৳ per sack

7 · Dadon: from whom, ৳

7 · Had to sell to lender
☐ Yes ☐ No

8 · Later price ৳/kg, month

## C · COLD STORAGE

9 · Sacks stored · table / seed

10 · Kg in a sack

10 · Rent ৳ per sack / per kg

10 · Same rent for 1 or 9 months
☐ Yes ☐ No ☐ Don't know

11 · Empty sack ৳

11 · Carry in ৳/sack · load-unload ৳/sack

12 · Advance ৳ per sack

12 · Interest or charge · repay by · if not repaid

13 · Month taken out

13 · Kg per sack in → out

13 · Kg rotten / sorted out

14 · Sale from store ৳/sack

15 · Why then / 16 · why not stored

## D, E, F · HOME STORAGE, MONEY, GROUPS

17 · Keeps at home (months)

17 · Kg lost per maund a month

18 · Cash need: for what, ৳, days

19 · Bank / NGO loan, interest %

20 · Would take a loan on sacks · ৳
☐ Yes ☐ No ☐ Maybe

21 · Group storage / shared truck
☐ Yes ☐ No ☐ Maybe

## G · THE CALL

22 · Repeated the advice correctly
☐ Yes ☐ Partly ☐ No

23 · Prefers · trusts it
☐ Keys ☐ Voice ☐ Trusts ☐ Not sure

24 · Would have sold later if knew

25 · Exact words (the one problem, or any strong quote)

ANSWER FORM · PRINT ONE PER COLD STORAGE

# Cold storage হিমাগার C ____

Date

Upazila

Interviewers · role of the person

Consent given
☐ Yes ☐ No

Rate card photographed
☐ Yes ☐ No

Told "no partners, demo only"
☐ Yes

## CAPACITY AND RENT

C1 · Capacity (sacks) · filled this season

C1 · Fills in (month) · empties by

C2 · Kg in a sack

C2 · Rent ৳ per sack / per kg

C2 · Same for 1 or 9 months
☐ Yes ☐ No

C2 · Rent paid when

C3 · Discounts: from how many sacks · how much off · for groups?

## ADVANCE, LOSSES, OTHER CHARGES

C4 · Advance ৳ per sack

C4 · Interest / service charge

C4 · If not repaid

C5 · Weight loss % a month

C5 · Rotten / sorted out % at exit

C6 · Loading, sacks, stacking ৳

## RECEIPTS, WHO STORES, BOOKINGS

C7 · Receipt given

C7 · Would confirm sacks to a bank
☐ Yes ☐ No ☐ Maybe

C7 · Banks already lend on receipts
☐ Yes ☐ No

C8 · Farmers % · traders %

C8 · Table % · seed %

C9 · Farmers take out in · traders in

C10 · Value of early group bookings · would pay a fee per booking?

Exact words worth quoting

## Tally sheet

Fill after the visits, one row per farmer. Write the unit with every number.

| CODE | UPAZILA | SOLD: MONTH, ৳/KG | WHY (4) | STORED (SACKS) | KG / SACK | RENT ৳/SACK | ADVANCE ৳/SACK, INTEREST | KG IN → OUT | TAKEN OUT (MONTH) | REPEATED ADVICE (Y/P/N) |
|---|---|---|---|---|---|---|---|---|---|---|
| F01 | | | | | | | | | | |
| F02 | | | | | | | | | | |
| F03 | | | | | | | | | | |
| F04 | | | | | | | | | | |
| F05 | | | | | | | | | | |
| F06 | | | | | | | | | | |
| F07 | | | | | | | | | | |
| F08 | | | | | | | | | | |
| F09 | | | | | | | | | | |
| F10 | | | | | | | | | | |

## Update the engine

Write the **middle value** of what you heard, the **range**, and **how many** people said it. Then give this page to whoever updates `krishivault-pipeline/src/engine/facts.py` (each value's source is written there too). To try a value before it is changed, edit the costs on the dashboard's "Store or sell" page.

| NUMBER | IN THE ENGINE NOW | WHERE IT CAME FROM | FIELD RESULT (MIDDLE, RANGE, HOW MANY) |
|---|---|---|---|
| Kg in a cold-storage sack | 84 kg | Field notes | |
| Rent for the season | ৳250 a sack, the same for 1 to 9 months | Field notes (some ৳300; some charge per kg) | |
| Empty sack | ৳40 | Guess | |
| Carrying to the cold storage | ৳25 a sack | Guess | |
| Loading and unloading | ৳15 a sack | Guess | |
| Selling from the cold storage | ৳30 a sack | Guess | |
| Selling at the field | ৳10 a sack | Guess | |
| Weight loss in cold storage | 0.5% a month | Guess | |
| Sorted out as rotten at the exit | 2% | Guess | |
| Loss in a home store | 4% a month, at most 3 months | Guess (3 months: field notes) | |
| Cold-storage advance | ৳250 a sack | Field notes (৳200-300) | |
| Interest on the advance | 14% a year | Guess | |
| Bank or NGO loan interest | 11% a year | Guess | |
| Farm-gate price, month by month | Moves with town retail prices | Assumption | |

## Turning real answers into slides

Fill these only with what you collected. Write the sample size every time ("of 7 farmers").

> **Voices from Bogura.** Two or three short quotes in Bangla with an English line, signed "farmer, Sariakandi" (no name unless they agree).

> **When farmers sell.** "___ of ___ farmers sold within 3 months of the harvest; the main reason was ___."

> **The flat rent.** "Both cold storages charge ৳___ a sack (___ kg) for the season, the same for 1 month or 9."

> **The price farmers missed.** "They sold at ৳___/kg in [month]; the market reached ৳___/kg by [month]."

> **The advance.** "Cold storages lend ৳___ a sack at ___%; ___ of ___ stored farmers took it."

> **Losses.** "Stored potatoes lost about ___% a month; ___% was sorted out at the exit."

> **Loans against sacks.** "___ of ___ farmers would take a bank loan against stored sacks; ___ of 2 cold storages would confirm the sacks to a bank."

> **The call works.** "___ of ___ farmers repeated our call's advice correctly after hearing it once."

## What we expect to hear: prepare, don't quote

These are the team's own field notes (14 October 2026). Use them to know what to ask more about, **never as answers from this field check**.

- The cold-storage sack is 84 kg; rent is about ৳250 a sack (some ৳300) for up to 9 months, the same for any length of stay. Some cold storages charge by the kg.
- A cold storage holds 50,000 to 2.5 lakh sacks.
- Cold storages lend ৳200 to ৳300 a sack; the stored sacks are the security.
- Potato keeps about 3 months at home.
- Farmers who store tend to sell within 1 to 3 months, afraid the price will fall; traders store potatoes for sale and release them when prices rise. Farmers mostly store only seed potato.
- Farmers sell retail (shops and people they know) and at the haat, where traders buy in bulk.

**If an answer surprises you** (a rent charged per kg, a different sack size, no flat rent), write it exactly and ask a follow-up. A surprise is the most useful thing this check can find: it changes the engine's sums.